

```
!pip install transformers
!pip install torch
!pip install wordcloud
```

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Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from wordcloud import WordCloud

from transformers import pipeline
```

```
generator = pipeline(
    'text-generation',
    model='gpt2'
)

print("GPT-2 Model Loaded Successfully!")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public model
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to
config.json: 100% 665/665 [00:00<00:00, 58.0kB/s]

model.safetensors: 100% 548M/548M [00:08<00:00, 75.1MB/s]

Loading weights: 100% 148/148 [00:00<00:00, 487.13it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----+-----+--
h.{0...11}.attn.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you
generation_config.json: 100% 124/124 [00:00<00:00, 11.9kB/s]

tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 2.12kB/s]

vocab.json: 1.04M/? [00:00<00:00, 15.4MB/s]

merges.txt: 456k/? [00:00<00:00, 2.15MB/s]

tokenizer.json: 1.36M/? [00:00<00:00, 2.87MB/s]
GPT-2 Model Loaded Successfully!

```

```
prompt = "Artificial Intelligence will"
```

```

output = generator(
    prompt,
    max_length=50,
    num_return_sequences=1
)

```

```
print(output[0]['generated_text'])
```

Passing `generation\_config` together with generation-related arguments=({'num_return_sequences': 1, 'max_length': 50, 'max_new_tokens': 256, 'pad_token_id': 50256})
 Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
 Both `max\_new\_tokens` (=256) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` is ignored when `max\_length` is not None.
 Artificial Intelligence will be able to do everything, from designing every vehicle to do

The challenge is that artificial intelligence needs to be intelligent enough to perform

These are some of the challenges AI will face, including:

Making a living

Running businesses

Driving cars

Using technology that is not used by humans

Making sure everything is safe

Making sure everyone has access to the right technology

Making sure the right people are available

Making sure everyone's jobs are safe

Making sure all the people working in the world are safe

Making sure everyone's pensions are good

Making sure all of the people who work in industries like IT or the military are safe

Making sure everyone's jobs are safe and all of the people working in the world are safe

So what does this mean for the future of AI?

This is where AI is coming in. In fact, the big question is where AI will come from. The

As AI enters the market, companies are going to use it to solve

```
prompts = [
    "The future of education is",
    "Once upon a time",
    "Machine learning helps",
    "Healthcare systems can improve by",
    "Climate change can be reduced by"
]

responses = []

for prompt in prompts:
    result = generator(
        prompt,
        max_length=60,
        num_return_sequences=1
    )

    generated_text = result[0]['generated_text']

    responses.append(generated_text)

    print("\n=====")
    print("PROMPT:")
    print(prompt)

    print("\nGENERATED RESPONSE:")
    print(generated_text)
```

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
Both `max\_new\_tokens` (=256) and `max\_length` (=60) seem to have been set. `max\_new\_tokens`
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
Both `max\_new\_tokens` (=256) and `max\_length` (=60) seem to have been set. `max\_new\_tokens`

=====

PROMPT:

The future of education is

GENERATED RESPONSE:

The future of education is still unknown", says the report.

"It is clear that the present educational system is failing to maintain, promote and create

The report calls for a greater emphasis on critical thinking and critical thinking is a

The report states: "Although we have many opportunities to support the growth and development

The report states: "We need to ensure that a student is well supported in the classroom

The report also calls for the development of a new system for teaching "where students

The report concludes: "

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Both `max\_new\_tokens` (=256) and `max\_length` (=60) seem to have been set. `max\_new\_tokens`

=====

PROMPT:

Once upon a time

GENERATED RESPONSE:

Once upon a time, we are told that the world is a simulation. But we are in fact merely

The illusion is one of the most obvious, and perhaps most misunderstood, characteristic

If you consider the fact that there is nothing more real than the world, and that its e

It is a simple act to use this illusion to create the illusion of reality. You do not l
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.

Both 'max_new_tokens' (=256) and 'max_length' (=60) seem to have been set. 'max_new_toks

=====

PROMPT:

Machine learning helps

GENERATED RESPONSE:

Machine learning helps us to understand the world around us. As far as we can tell, it

In this article, we will use a simple machine learning approach to determine if a perso

What happens when a person's behaviour changes

In a simple form, a person's behaviour changes by changing their brain activity. The ne

The brain activity of the brain is thought to be involved in learning. The brain is the

Learning

```
print("==== CONTEXT ANALYSIS ====")
```

```
for i in range(len(prompts)):
```

```
    print("\nPrompt:", prompts[i])
```

```
        print("Observation:")
```

```
        print("- GPT-2 generated context-related text.")
```

```
        print("- Response maintained topic continuity.")
```

```
        print("- Model showed language understanding ability.")
```

==== CONTEXT ANALYSIS =====

Prompt: The future of education is

Observation:

- GPT-2 generated context-related text.
- Response maintained topic continuity.
- Model showed language understanding ability.

Prompt: Once upon a time

Observation:

- GPT-2 generated context-related text.
- Response maintained topic continuity.
- Model showed language understanding ability.

Prompt: Machine learning helps

Observation:

- GPT-2 generated context-related text.
- Response maintained topic continuity.
- Model showed language understanding ability.

Prompt: Healthcare systems can improve by

Observation:

- GPT-2 generated context-related text.
- Response maintained topic continuity.
- Model showed language understanding ability.

Prompt: Climate change can be reduced by

Observation:

- GPT-2 generated context-related text.
- Response maintained topic continuity.
- Model showed language understanding ability.

```
print("==== LANGUAGE GENERATION ANALYSIS ====")

print("""
Observations:

1. GPT-2 generates human-like sentences.
2. The model predicts next words using learned patterns.
3. Responses are context-aware.
4. Some outputs may become repetitive.
5. Creativity depends on prompt quality.
""")
```

```
==== LANGUAGE GENERATION ANALYSIS ====

Observations:

1. GPT-2 generates human-like sentences.
2. The model predicts next words using learned patterns.
3. Responses are context-aware.
4. Some outputs may become repetitive.
5. Creativity depends on prompt quality.
```

```
research_questions = [
    "How effectively does GPT-2 understand contextual prompts?",
    "Can GPT-2 generate coherent text for different domains?",
    "Does GPT-2 maintain topic consistency?",
    "How creative are GPT-2 generated responses?",
    "What are the limitations of GPT-2?"
]

print("==== RESEARCH QUESTIONS ====")

for q in research_questions:
    print("-", q)
```

```
==== RESEARCH QUESTIONS ====
- How effectively does GPT-2 understand contextual prompts?
- Can GPT-2 generate coherent text for different domains?
- Does GPT-2 maintain topic consistency?
- How creative are GPT-2 generated responses?
- What are the limitations of GPT-2?
```

```
prompt_lengths = [len(p.split()) for p in prompts]
response_lengths = [len(r.split()) for r in responses]

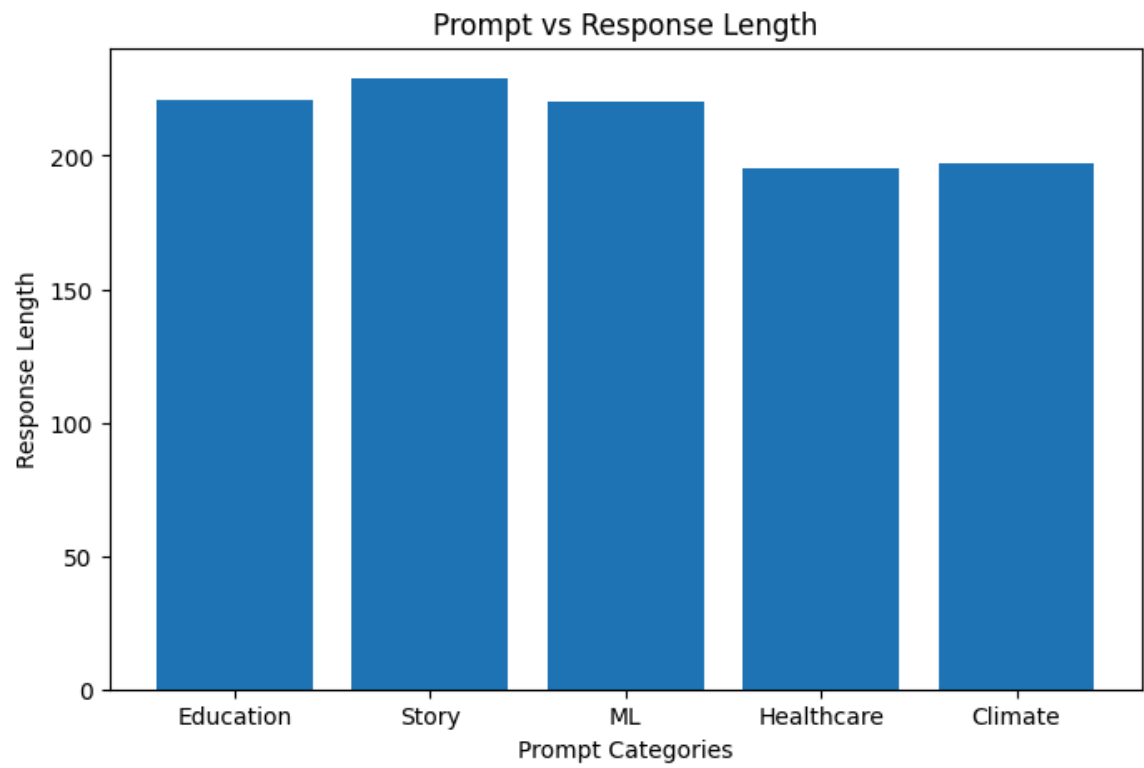
plt.figure(figsize=(8,5))

plt.bar(range(len(prompts)), response_lengths)

plt.xticks(range(len(prompts)), [
    "Education",
    "Story",
    "ML",
    "Healthcare",
    "Climate"
])

plt.xlabel("Prompt Categories")
plt.ylabel("Response Length")
plt.title("Prompt vs Response Length")

plt.show()
```



```
all_text = " ".join(responses)

wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='white'
).generate(all_text)

plt.figure(figsize=(10,5))

plt.imshow(wordcloud)

plt.axis('off')

plt.title("Word Cloud of GPT-2 Responses")

plt.show()
```



```

- AI-generated text may contain misinformation.
- Language models should be used responsibly.
- Human verification is important.
- Bias in training data may affect outputs.
"""
)

```


Ethical Considerations:

- AI-generated text may contain misinformation.
- Language models should be used responsibly.
- Human verification is important.
- Bias in training data may affect outputs.

```
print("""
===== CONCLUSION =====
```

This project explored the GPT-2 Language Model using NLP techniques.

The model demonstrated:

- Context understanding
- Human-like text generation
- Creativity in responses
- Adaptability across domains

The analysis showed that GPT-2 performs effectively for text generation tasks, although

This project provided practical understanding of Language Models, NLP, and AI-driven te